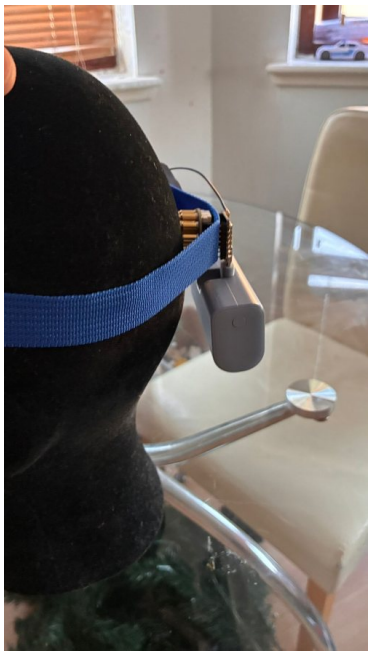
QUICK START & SAFETY GUIDE · SKU PIA24027P

From the Box to 16 Live Channels

Seven steps: mount the XIAO, flash the firmware from the browser, power from a power bank, connect over Bluetooth, put it on and check the signal. Plus pogo-pin care and troubleshooting.



WARNING Power bank only while worn.

Octopus 16 has no isolation from its power supply. While it is on a head it runs from a small 5 V power bank only — never plugged into a computer or a wall charger. USB from a computer is for flashing only, with the board off the head. Octopus 16 is not a medical device.

01 SAFETY RULES

#	Rule
1	**Power bank only while worn.** Never plugged into a computer or wall charger while on a head.
2	**Flash off the head.** USB from a computer is only for flashing.
3	**No diagnosis.** Band power, “focus” and “relax” ratios are uncalibrated learning indicators.
4	**Implanted devices.** Not on a person with a pacemaker or implant without a physician’s approval.
5	**Skin.** Clean, intact skin. Stop if irritated or the pins press too hard.
6	**Pin care.** Press straight down; clean with isopropyl alcohol, never oil or grease.

Power source	While worn	Why
Small 5 V power bank	SAFE	On its own battery, no path to the mains.
Laptop USB	NEVER	USB from a computer is for flashing only.
Wall charger	NEVER	Connects the person to the mains.

02 WHAT YOU NEED

Item	Requirement
Octopus 16	Included
XIAO ESP32-S3 + BLE antenna	Included in the Elecrow package; headers soldered
Small 5 V power bank	≤ 2 A
USB-C data cable	Must carry data, for flashing
Strap or headband	Elastic band or 3D-printed holder
Bluetooth LE device	Chrome or Edge (not Safari, Firefox or iOS), or Python 3.10+

03 SETUP IN SEVEN STEPS

1 Gather the parts

Octopus 16, XIAO ESP32-S3 with antenna, USB-C data cable, power bank, strap.

2 Mount the XIAO the right way round

Line up the **5V, GND, 3V3** labels on both boards, press down gently and clip on the antenna.

3 Flash the firmware (once)

Board **off the head**. Open **octobridge.pieeg.com/flash** in desktop Chrome or Edge, plug in the XIAO, click **Install Octopus-16**, wait ~30 s, replug. No port? Hold BOOT, tap RESET, release BOOT, retry.

4 Power from the power bank

Unplug the computer and connect the power bank to the XIAO USB-C. The LED blinks slowly.

5 Connect over Bluetooth

cloud.pieeg.com → Session Lobby → PiEEG Devices → Octopus 16, or **octobridge.pieeg.com/connect**, then pick Octopus-16-XXXX. The LED turns solid.

6 Put it on

Over **Pz** for the alpha test. Part the hair, press straight down, hold with the strap; the power bank sits under the strap.

7 Check the signal

"EEG ok" green (RMS 2–200 µV). Close the eyes 10 s: alpha rises. Blink over Fp: large spikes. Jaw clench: RMS and gamma jump.

04 LED STATES

Pattern	State
Solid (brief)	Booting
Slow blink, ~1 s	Advertising, waiting for an app
Solid	Connected and streaming
Fast double-blink	ADC not detected — XIAO mounted the wrong way round

05 TROUBLESHOOTING

Flashing and Bluetooth

Problem	Fix
No serial port	Use a USB-C data cable; close other serial programs.
Install fails	BOOT + RESET, then Install again.
Octopus-16-XXXX not listed	Power bank on, LED slow-blinking, move closer; flash if never flashed.
Other name (Octopos, FE-Gamepad-...)	Different firmware build; reflash, or connect by service UUID.
Connects then drops	Antenna clipped on, board close, retry.
No Bluetooth button	Use Chrome or Edge.

Data and signal

Problem	Fix
No EEG, LED double-blinks	XIAO is 180° wrong — reseat with labels aligned.
Gaps in data	Lost packets: check the counter, move closer, avoid busy 2.4 GHz.
"EEG ok" red	Press straight down, part the hair, sit still.
A few channels noisy	Those pins are not touching or are stuck.
Strong 50/60 Hz hum	Something is on mains power — power bank only.

06 POGO-PIN CARE

- Clean pin tips with isopropyl alcohol > 90 % on a swab.
- Blow dust out of the pin bases with compressed air.
- Press straight down (90°); never tilt or push sideways under pressure.
- Do not over-compress the pins.

Free a stuck pin: a drop of IPA into the barrel, gently work the plunger with a non-metal tool, let it dry fully before powering on.

WARNING Never oil or grease.

It attracts dirt and ruins the contact. Damage from dirt or rough handling is not covered by the warranty.

07 RELATED DOCUMENTS

OC16-DS-001 Product Datasheet	**OC16-MD-002** Mechanical & Dimensional Drawing	**OC16-PCB-003** PCB & Manufacturing Specification
OC16-PIN-004 Electrode Array & Pinout Guide	**OC16-SW-006** Bluetooth Interface & Software Guide	